Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

Table of contents

Approvals 4
Revision history 5
Abbreviations 5

1 Purpose and scope **7**
1.1 Changes in this revision 7

2 Objectives **11**

3 Related documents **12**

4 Prior knowledge and quality risk basis **14**
4.1 Unit-operation description and prior knowledge 14
4.2 Quality attributes in scope 16
4.3 Risk-based prioritization of parameters 17

5 Materials and methods **19**
5.1 Scale-down model and its qualification 19
5.2 Cell line, media and culture operation 20
5.3 Analytical methods 21
5.4 Sampling plan 22
5.5 Statistical methods 22

6 Study design **24**
6.1 Factors, ranges and study type 24
6.2 Screening design 25
6.3 Response-surface design 25
6.4 Univariate assessment 26

7 Acceptance and decision criteria **27**

8 Proven acceptable ranges (planned analysis) **32**

9 Data management and integrity **34**

10 Roles and responsibilities **35**

11 Deliverables and schedule **36**

12 Risks and assumptions **37**

13 Approvals **39**

Appendix A. Planned screening design **40**

Appendix B. Planned response-surface design **41**

References **42**

Field	Value
Document ID	PCP-003
Document class	Process Characterization Plan
Title	Process Characterization Plan: Production Bioreactor (Step 3)
Product	A-Mab
Version	2.0
Effective date	2026-09-30
Supersedes	PCP-003 Version 1.0 (2025-08-07)
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

> **SYNTHETIC DOCUMENT — NOT A REAL RECORD.** Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-08-07	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
%Gal	total galactosylation, as a percentage of the glycans
2-AB	2-aminobenzamide
ALCOA+	attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CEX	cation exchange chromatography
CHO	Chinese hamster ovary
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DoE	design of experiments
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
FcγRIIIa	Fc gamma receptor IIIa
G1, G2	glycans with one or two terminal galactose residues
GDP	guanosine diphosphate
GPP	general process parameter
H	high (criticality level)
HCP	host cell protein
HILIC	hydrophilic interaction liquid chromatography
HMW	high molecular weight species
HPLC	high-performance liquid chromatography
IAC	intermediate acceptance criterion
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G, subclass 1
KPP	key process parameter
LMW	low molecular weight species
M-H	moderate to high (criticality level)

Abbreviation	Meaning
Man5	oligomannose glycan with five mannose residues
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
pCO2	partial pressure of dissolved carbon dioxide
PPQ	process performance qualification
PRESS	prediction error sum of squares
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
sd	standard deviation
SEC	size exclusion chromatography
SOP	standard operating procedure
UDP	uridine diphosphate
UPLC	ultra-performance liquid chromatography
VH	very high (criticality level)
VL	very low (criticality level)
WC-CPP	well-controlled critical process parameter

1 Purpose and scope

This plan describes the Stage 1 characterization of the production bioreactor, Step 3 of the A-Mab drug substance process (U.S. Food and Drug Administration 2011). The production bioreactor is a fed-batch culture with a working volume of 15,000 L at commercial scale. The step forms the glycan, charge variant and aggregate quality attributes of A-Mab (Section 4.1). The plan applies the strategy, the statistical approach and the framework of acceptance criteria of the Process Characterization Master Plan PCMP-001 to this step, following the approach of the A-Mab case study (CMC Biotech Working Group 2009).

The studies are run in a scale-down model of the production bioreactor, whose qualification against manufacturing scale is planned in Section 5.1. They cover the 9 process parameters of the step that the Pre-Characterization Process Risk Assessment RA-001 brought into scope (Table 6.1). They also cover the 7 quality attributes that the drug substance CQA register assigns to the step. Of these, 5 are responses of the designs, and host cell protein and residual DNA are measured on the clarified harvest of every run (Section 4.2). The plan states the criteria against which each result will be judged and the analysis from which the proven acceptable ranges (PARs) of the step will be derived. The results will be reported in the paired Process Characterization Report PCR-003, which rolls up into the Process Characterization Master Report PCMR-001.

The characterization supports the transfer of the process from the development site in Cambridge, MA, to the commercial drug substance facility in Grafton, WI, under the Process Transfer Plan PTP-001. The results are the evidence for the operating ranges of the production bioreactor and for the part of the control strategy that this step carries.

Some related work lies outside the scope of this plan. The seed expansion steps that precede the production bioreactor are not characterized here. Harvest and clarification (Step 4) is characterized under its own plan, PCP-004. Fragments are not an attribute of the bioreactor studies (Section 4.2). Confirmation of the ranges at commercial scale belongs to Stage 2 and will be made during process performance qualification (PPQ) at the receiving site (U.S. Food and Drug Administration 2011). The studies planned here are Stage 1 process design studies, and the claims they support are bounded by the scale-down model on which they are run.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-08-07, before the first study of the step, the qualification study S-BRX-SDM, started on 2025-08-11. Version 2.0 is the protocol

amendment that the QA investigation INV-001 required in its corrective action CA2, which amends every step plan with the new criteria and the method for fragments. QA opened the investigation INV-001 on 2026-08-03 after its review of the master report PCMR-001 Version 1.0 for BLA readiness. QA closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the proven acceptable range analysis were in no approved protocol and had been set after the data were seen (INV-001 finding 1). The limits that the Version 1.0 report PCR-003 applied to the outputs of this step were among them. This revision makes the following changes, each set by the master plan PCMP-001 Version 2.0.

1. Version 2.0 states an intermediate acceptance criterion (IAC) for each of the 5 responses of the step (Section 7, Table 7.1). The master plan PCMP-001 derives these criteria a priori from the drug substance criteria and the worst case of every downstream step on the platform (PCMP-001 §7.3). They replace the criterion basis of Version 1.0, an in-process limit derived from the nominal downstream clearance with a margin held between the two. That basis was written into the plan text when the package was compiled, and it was not approved with the plan (INV-001 finding 1). They also replace the Version 1.0 analysis limits. For the glycans and acidic variants, the Version 1.0 upper limit was the mean plus 2.5 standard deviations of the capability data that it was then used to judge (INV-001 finding 3). For aggregate, Version 1.0 carried the drug substance limit back through the nominal downstream clearance of this campaign and then divided it by a safety factor of 6.8 (INV-001 finding 2).
2. Version 2.0 states the rule by which the verification of the platform worst cases may tighten or widen a criterion (PCMP-001 §7.5). That rule changed the criterion of this step for aggregate, which was tightened and then widened (Section 7).
3. Version 2.0 judges acidic variants against both the lower and the upper limit of their criterion (INV-001 finding 4). Capability is reported as the lower of the two one-sided indices for each criterion with two limits, where Version 1.0 reported a one-sided Cpk. Capability is judged against a Cpk of 1.33, expected in routine manufacture, and a Cpk of 1, below which an action is needed. Version 1.0 set no capability threshold.
4. Version 2.0 qualifies the scale-down model for each output by its offset from manufacturing scale, and it states every judgement against a criterion at manufacturing scale (Section 5.1, PCMP-001 §7.1). This replaces the Version 1.0 criterion of no statistically significant difference between the scales at $p < 0.05$, and the Version 1.0 requirement to investigate a difference before any characterization run. Version 2.0 names the runs and batches compared. The decision whether the model is qualified for an output is made at analysis.
5. Version 2.0 states the campaign thresholds for model adequacy (PCMP-001 §7.2). The model F-test must be significant at $\alpha = 0.05$, lack of fit must not be significant at $\alpha = 0.05$, and predicted R^2 must fall short of R^2 by no more than 0.20. The last replaces the Version 1.0 requirement that predicted R^2 track adjusted R^2. The remedy of PCMP-001 §7.2 replaces the Version 1.0 consequence of a failing model, which was that the model would be reduced, or the design augmented, and the revised model reported with the reason for the revision. A model that fails the lack-of-fit test or the limit on overfitting is now either refitted with a

justified change of terms or transformation, or it is not used to set a range. The other adequacy rules of Version 1.0 are kept.

6. Version 2.0 states the tests by which the residuals of a response-surface model are judged (Section 7). A regression of the absolute residuals on the predicted values, and a regression of the residuals on run order, each show no slope significant at $\alpha = 0.05$. A run with a Cook's distance above 4/n is examined by deleting it and repeating the judgement. The model fails this test only where a run's Cook's distance exceeds the median of the F distribution with p and n − p degrees of freedom, where p is the number of model parameters and n the number of runs.
7. Version 2.0 judges each quality attribute in the PAR analysis on the 95 % prediction interval of a single batch, which includes the model residual and the uncertainty of the fitted coefficients (PCMP-001 §6.4). The other factors are drawn from a normal distribution about their set-points and clipped to the range studied (Section 8). Version 1.0 drew them from within their NORs and gave no interval.
8. Version 2.0 draws the operating region, which PCMP-001 calls the design space, on the single-batch limit (PCMP-001 §6.2). A setting lies inside the region when every modelled attribute meets its criterion at both ends of the 95 % prediction interval of a single run (Section 7). Version 1.0 declared the region where the response-surface models predict that every modelled attribute meets its criterion, that is on the mean prediction.
9. Version 2.0 adds two rules of PCMP-001 to the classification, whose Version 1.0 rule is kept word for word (Section 7). Under PCMP-001 §6.2 an attribute is affected by a parameter when a term that contains the parameter is significant. Under PCMP-001 §7.7 a significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical. The report states the size of each effect across the PAR and the margin of the PAR to the NOR, and it argues each class on them. Version 1.0 did not define when a parameter affects an attribute, and it classified every parameter that affects a critical quality attribute as critical or as well controlled critical.
10. Version 2.0 states the grid of the analysis at the set-points as the 201 points that the analysis uses. Version 1.0 stated 81 points for both analyses.
11. Version 2.0 records the execution window of each study and the calibration status of the pCO2 probe that the studies used (Section 11). The Version 1.0 records gave no execution dates (INV-001 finding 5). The Version 1.0 title block carried 2026-07-24, the date on which the Version 1.0 reports were approved, in place of the date on which this plan was approved (INV-001 finding 5). The compiled text of Version 1.0 also carried the criterion basis that change 1 replaces, added to the plan text when the package was compiled, without a revision of the approved plan (INV-001 finding 1). The revision history now gives the approval date.
12. Version 2.0 gives the parameter names and units of the current configuration. The pCO2 parameter is named CO2 partial pressure (pCO2), and the units are given in full for osmolality (mOsm/kg), dissolved oxygen (% air saturation), initial viable cell concentration (10^6 cells/mL), basal medium concentration (× standard) and nutrient feed-1 volume (% of working volume) (Table 6.1). No value changes.
13. Version 2.0 corrects the description of high mannose glycans in Section 4.1. They

are the glycans whose processing stops before N-acetylglucosaminyltransferase I, not glycans on which processing never began.

14. Version 2.0 corrects the description of the control of culture pH in Section 5.2. Version 1.0 named carbon dioxide addition and base addition. Carbon dioxide cannot set the pH where pCO2 is a factor of the designs, so culture pH is held by base addition and pCO2 by the carbon dioxide fraction of the sparge gas.

The factors, their ranges and set-points, the designs, the sampling and the other decision rules are those of Version 1.0. The revision history also names INV-002, the investigation of the response-surface models that did not meet the model adequacy criteria of PCMP-001. The investigation INV-002 changes nothing in this plan. What it changed in the analysis of the bioreactor studies is recorded in the bioreactor report PCR-003 as discrepancies from this plan.

The analytical method decision AMD-001 of 2026-03-20 made non-reduced CE-SDS (nrCE-SDS) the method for fragments at every step. The decision to measure fragments by nrCE-SDS does not change the methods of this plan, because fragments are not an attribute of the bioreactor studies (Section 4.2). PCMP-001 keeps SEC as the method for aggregate (PCMP-001 §7.8).

Table 1.1 sets the Version 1.0 limits beside the criteria now in force. Every upper limit in Version 2.0 is higher than the limit that Version 1.0 applied, and Version 2.0 adds a lower limit of 18.0 % for acidic variants. The Version 1.0 lower limits of afucosylation, galactosylation and high mannose were their drug substance lower limits, and they are unchanged. The criteria in force are derived without data from the bioreactor studies.

Table 1.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, production bioreactor.

Attribute	Version 1.0[a]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Acidic variants (%)	≤ 25.8	R2	18.0–39.0	Carried back
Afucosylation (%)	2.00–9.43	R2	2.00–13.0	Drug substance
Aggregates, HMW (%)	≤ 1.86	R1, margin 6.8	≤ 6.46	Carried back, tightened[b] and widened[c] from ≤ 9.30
Galactosylation (%)	10.0–30.1	R2	10.0–40.0	Drug substance
High mannose (%)	3.00–7.28	R2	3.00–10.0	Drug substance

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[c] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[R1] Version 1.0 rule: nominal back-calculation / margin.
[R2] Version 1.0 rule: mean + 2.5 sd of capability.

2 Objectives

The study has the following objectives.

1. To quantify the relationship between each studied parameter and each of the 5 modelled responses over the ranges studied in Table 6.1, and to identify the interactions between parameters.
2. To define the multivariate region of the response-surface parameters over which every modelled attribute is predicted to meet its intermediate acceptance criterion.
3. To establish a proven acceptable range for each parameter against each attribute whose fitted model contains a significant term in that parameter, and to confirm that the NOR in Table 6.1 lies inside that range.
4. To provide the demonstrated effects and the operating risk from which each parameter is classified under the parameter classification procedure SOP-4001.
5. To provide the process understanding and the capability estimate that justify the Stage 2 operating ranges and the contribution of this step to the control strategy.

3 Related documents

Table 3.1 lists the parent documents of this plan, its paired report, the master report and the investigation INV-001. This plan also cites the harvest plan PCP-004, the Protein A, cation exchange and anion exchange reports PCR-005, PCR-007 and PCR-008, the low-pH viral inactivation report PCR-006, the analytical method decision AMD-001 and the investigation INV-002, each where it is used.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

Table 3.2 lists the procedures and method validations that this plan cites. The SOP and AMV numbers are placeholders.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2003	Operation of the Production Bioreactor (Fed-Batch)
	SOP-2004	Cell-Culture Media and Feed Preparation
	SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3012	Host-Cell-Protein Quantitation by ELISA
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3014	Residual Host-Cell DNA by qPCR
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3013	Charge Variants (icIEF)
	AMV-3014	Residual DNA (qPCR)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch culture in which a recombinant Chinese hamster ovary (CHO) cell line secretes A-Mab into the medium over a culture of about 17 days. The pH, temperature, dissolved oxygen, pCO2 and osmolality of the culture are held at the levels of the run throughout that period. Each of them changes the specific growth rate of the cells and the rates of the Golgi enzymes that complete the processing of the antibody. The harvested pool contains everything the culture secreted during the run. A longer culture therefore weights the pool toward antibody made after the nucleotide sugar donors have been drawn down and after viability has begun to fall. The output of the step is the culture fluid at the end of the run, which passes to harvest and clarification (Step 4).

The platform on which A-Mab is produced was used for the humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab and for the A-Mab clinical and engineering campaigns (CMC Biotech Working Group 2009). Cell line construction, basal medium, feed strategy and the control of pH, temperature and dissolved oxygen are common to those products. The same host cell line carries the same glycosyltransferases and glycosidases, and the same medium and feed supply the same nucleotide sugar precursors. A parameter that lowered galactosylation in those products is therefore expected to lower it in A-Mab, through the same enzymes. The directions recorded on the platform are carried into this study as expectations, and the magnitudes are not.

Glycosylation is completed in the Golgi before the antibody is secreted. Protein A binds the Fc region outside the glycan, and the cation and anion exchange steps separate by net charge. Core fucose, galactose and mannose carry no charge, so none of the three purification steps retains one neutral glycoform in preference to another (CMC Biotech Working Group 2009). The distribution of the neutral glycoforms in the drug substance is therefore decided in the bioreactor. Sialic acid does carry a charge, and sialylated glycans are counted among the acidic variants. Aggregate and acidic variants are also formed in the culture, and the purification train can reduce that burden but cannot reverse it.

Core fucose is added by a fucosyltransferase from the donor GDP-fucose. The fraction of antibody that leaves the cell without core fucose therefore rises when the activity of that enzyme falls, when GDP-fucose is depleted, or when the glycan spends less time in the Golgi before it is secreted. Terminal galactose is added by a galactosyltransferase from the donor UDP-galactose, and both that donor and the manganese cofactor of the enzyme are depleted late in a fed-batch culture. Galactosylation is therefore expected to fall as the culture ages and the harvested pool accumulates

late-culture antibody. High mannose glycans, mainly Man5, remain when processing stops before N-acetylglucosaminyltransferase I adds the first N-acetylglucosamine. Their level is expected to rise when transit through the Golgi is fast relative to the processing enzymes, or when those enzymes are inhibited.

The fucosyltransferase, the galactosyltransferase and the mannosidases draw on nucleotide sugar pools of limited size and work at the same Golgi luminal pH. Raising culture pH or culture temperature therefore changes the rate of all of them at once. When a donor pool limits a reaction, a change that speeds the enzyme raises the product only as far as the donor allows. The amount by which one parameter moves a glycan attribute is therefore expected to depend on the levels of the others.

Aggregate forms after the antibody is secreted, while it sits in the culture fluid at culture temperature for days. A higher titer, a longer culture and a culture pH closer to the isoelectric point of the antibody each raise the aggregate level in the harvest. The isoelectric point lies above the culture pH. Raising culture pH across the range studied therefore lowers the net positive charge of the antibody, which weakens the electrostatic repulsion between molecules and raises the rate at which they associate.

Acidic variants arise mainly from asparagine deamidation, with smaller contributions from sialylated glycans and glycated lysines. The rate of deamidation rises with pH and with temperature, and the time available for it rises with the culture duration. Sialic acid is added by sialyltransferases inside the cell, and glycation adds glucose to lysine residues without an enzyme, at a rate set by the residual glucose in the medium. These three routes do not respond to a given parameter in the same direction. The net direction in which a parameter moves the acidic variant level is therefore not predictable from the chemistry alone.

Host cell protein and residual DNA enter the harvest when cells die and lyse. Viability falls late in a fed-batch culture, so a longer culture carries a larger amount of both into the harvest. Protein A capture, cation exchange and anion exchange remove most of both.

A higher culture pH raises the intracellular pH and with it the Golgi luminal pH. The glycosyltransferases and the mannosidases each work fastest at a different luminal pH, so a shift changes their rates by different amounts and moves the glycan distribution. A higher culture temperature raises the specific growth rate and the rate of every enzymatic and chemical reaction in and around the cell. A cooler culture grows more slowly and stays productive for longer, and a warmer one deamidates and aggregates faster in the culture fluid.

Dissolved carbon dioxide equilibrates with bicarbonate and lowers the cytosolic pH, so raising pCO2 moves the Golgi enzymes in the same direction as lowering culture pH. Holding the culture pH at its set-point against that acidification consumes more base, which raises the osmolality as bicarbonate and sodium accumulate. An osmolality above the set-point draws water out of the cell, slows the specific growth rate and raises the amount of antibody each surviving cell makes. It also changes the size of the nucleotide sugar pools that culture pH and culture temperature act on, so its effect on a glycan attribute is expected to depend on the levels of those two. A longer culture duration lowers viability at harvest and raises the fraction of the pool

secreted after the supply of UDP-galactose and manganese has fallen. It therefore lowers galactosylation and raises the amount of host cell protein and DNA carried into the harvest.

Dissolved oxygen supplies the oxygen for oxidative phosphorylation, so a lower level slows the rate at which the cell regenerates ATP. The culture is not limited by oxygen at either end of the range studied, so neither the specific growth rate nor the glycan distribution is expected to move measurably across it. A higher initial viable cell concentration brings the culture to its peak viable cell density earlier and leaves more days in the productive phase before harvest, which raises titer. A higher basal medium concentration and a larger nutrient feed-1 volume raise the supply of glucose, amino acids and nucleotide sugar precursors, which raises growth and titer. Across the ranges studied the feed replenishes those precursors faster than the cells consume them. None of these three parameters is therefore expected to move a glycan attribute at a fixed culture age.

4.2 Quality attributes in scope

Table 4.1 lists the 7 quality attributes that the drug substance CQA register assigns to the production bioreactor. Criticality was assigned in RA-001 with Tool #1 of the A-Mab case study, whose score is the product of the impact of the attribute on safety or efficacy and the uncertainty in that impact (CMC Biotech Working Group 2009). Criticality is treated as a continuum (International Council for Harmonisation 2009). The levels run from very low (VL) to very high (VH), and the attributes of high (H) and very high criticality are called critical. For this step the critical attributes are afucosylation, galactosylation, high mannose and aggregate. Host cell protein has moderate to high criticality (M-H). Acidic variants and residual DNA have very low criticality, and their drug substance criteria still apply (PCMP-001 §3).

Table 4.1: Quality attributes that the drug substance CQA register assigns to the production bioreactor, with their drug substance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36

CQA	Category	Acceptance	Criticality	Tool #1
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6

High mannose carries the highest Tool #1 score of the step. High mannose species are cleared from the circulation faster than the fully processed glycoforms, so a raised level shortens exposure. Afucosylation and galactosylation are scored high because A-Mab acts mainly through antibody-dependent cellular cytotoxicity. Antibody without core fucose binds more strongly to the Fc gamma receptor IIIa (FcγRIIIa) on the effector cell, and a higher affinity raises the cytotoxicity that the bound cell delivers. Aggregate is scored high for its immunogenic potential.

Of the 7 attributes, the 5 modelled responses of the designs are afucosylation, galactosylation, high mannose, acidic variants and aggregate. The register states the criterion for acidic variants for measurement by CEX or icIEF, and this plan measures them by icIEF (Section 5.3). Host cell protein and residual DNA will be measured on the clarified harvest of every run (Section 5.4), but they are not modelled responses of this study. Neither has an intermediate acceptance criterion at this step. Host cell protein is first judged in the Protein A pool, and residual DNA is judged on the drug substance only (PCMP-001 §7.3). The ranges over which their clearance is controlled are established in the Protein A, cation exchange and anion exchange reports PCR-005, PCR-007 and PCR-008.

Fragments do not appear in Table 4.1. Fragments are formed in the culture and during the hold of the harvest, by proteases released from lysed cells that clip the antibody. The CQA register assigns them to harvest. Fragments have intermediate acceptance criteria from the Protein A pool onward and none at this step (PCMP-001 §7.3). They are therefore not an attribute of the bioreactor studies.

4.3 Risk-based prioritization of parameters

The risk assessment RA-001 ranked every parameter of the step before characterization began. Each parameter was ranked on its potential impact on a quality attribute and on its potential to interact with other parameters (International Council for Harmonisation 2023). Parameters ranked high, or with a likely interaction, were assigned to multivariate study. Parameters ranked lower, whose range could be supported one factor at a time, were assigned to univariate study (PCMP-001 §4). The risk assessment RA-001 gives the scores and the rationale for each assignment, and they are not repeated here.

For the production bioreactor, 5 of the 9 parameters were assigned to multivariate study and 4 to univariate study (Table 6.1). The multivariate parameters are culture pH, culture temperature, CO2 partial pressure (pCO2), osmolality and culture duration. Each of them changes the Golgi luminal pH or the size of the nucleotide sugar pools, and the glycosyltransferases draw on both. The amount by which one of them moves a glycan attribute is therefore expected to depend on the levels of the

others, and only a design that varies them together estimates that dependence. The univariate parameters are dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume. Each of them raises or lowers the number of viable cells and the amount of antibody they make, and the feed holds the nucleotide sugar precursors in excess across the ranges studied. None of them is therefore expected to change the rate of a Golgi enzyme at a fixed culture age, and a study that varies one at a time supports the range each of them needs.

The range studied of every parameter extends beyond its NOR at both ends. The width of the range studied is between 1.5 and 2.5 times the width of the NOR. The wider range bounds the response of each attribute outside the region in which the process should run, which allows the severity of an excursion to be assessed. It also sets the knowledge space within which a PAR can be claimed. No PAR is claimed beyond the range studied.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run in bench-scale stirred-tank bioreactors of the same design class as the commercial production vessel, with the same control scheme for pH, temperature, dissolved oxygen and nutrient addition. Parameters that do not depend on scale, which include culture pH, culture temperature, dissolved oxygen, the initial viable cell concentration and the culture duration, are run at the commercial set-points in Table 6.1. Parameters that depend on scale, which include the agitation rate, the gas flow rates, the headspace pressure and the working volume, cannot be held equal across scales. They are set so that process performance at small scale matches performance at full scale (CMC Biotech Working Group 2009). The model is operated under the bioreactor operating procedure SOP-2003.

pCO2 is the parameter for which that distinction matters most. Carbon dioxide accumulates in a large vessel, because the depth of the liquid raises the hydrostatic pressure and lowers the volumetric mass transfer coefficient for carbon dioxide stripping. A bench-scale vessel strips carbon dioxide faster at the same gas flow per unit volume, so it reaches a lower pCO2 under the same nominal conditions. pCO2 is therefore controlled to the level of each run in the scale-down model and is studied as a factor in its own right, over a range that spans the levels the commercial vessel reaches. The carbon dioxide stripping of the commercial vessel is not matched, which departs from PCMP-001 §5 for this reason. The pCO2 of each run is measured by the pCO2 probe of the scale-down bioreactor system (EQ-BRX-205), which is calibrated and under change control (Section 9).

The qualification study S-BRX-SDM follows the scale-down model procedure SOP-1001. It compares 3 runs of the scale-down model at the set-points in Table 6.1 with 5 manufacturing-scale batches of the A-Mab engineering and clinical campaigns. The comparison covers the viable cell density profile, the viability at harvest, the titer and the 5 modelled responses, afucosylation, galactosylation, high mannose, acidic variants and aggregate. PCMP-001 requires the qualification runs to be made, and the offset of each output to be estimated, before the first characterization run of the step. The qualification study of this step ended before the screening study started (Table 11.1), and the report states when the offsets were estimated. A scale-down model need not reproduce the manufacturing mean. The difference between the two scales must instead be known precisely enough that no conclusion depends on its exact size. The model is qualified for each output separately, by the following rule (PCMP-001 §7.1).

1. The offset of an output is the mean of the scale-down runs minus the mean of the manufacturing-scale batches. It is reported with its two-sided 90 % confidence interval, computed from Student's t with the pooled standard deviation of the two scales. Each end of that interval is a one-sided 95 % confidence bound.
2. Every judgement of a scale-down result against a criterion first subtracts the offset of that output, in every study of this plan. Ranges and conclusions are therefore stated at manufacturing scale.
3. The conclusion for an output is whether it meets its criterion at all 81 combinations of NOR low, set-point and NOR high of the 4 response-surface factors. Each combination is judged on the exact 95 % prediction interval of a single run from the response-surface model.
4. The model is qualified for an output when this conclusion is the same with the offset at either end of its confidence interval.
5. An output with no criterion at the step, such as the viable cell density, the viability and the titer, has its offset recorded only.

The decision whether the model is qualified for an output is made at analysis, because the conclusion for the output needs the response-surface model. If the model is not qualified for an output, the report will state so and will state the consequence (PCMP-001 §5). The offset is estimated at the set-points only. Its use across the range studied is an assumption of this plan (Section 12).

The centre point runs of both designs are made at the same set-points as the qualification runs. They therefore carry the run-to-run variation of the model together with the variation of the analytical methods, and they give the pure error against which lack of fit is tested (Section 5.5).

5.2 Cell line, media and culture operation

Cells are thawed from the working cell bank and expanded through the seed train to the initial viable cell concentration of the run. The production bioreactor is operated in fed-batch mode under SOP-2003, with a single nutrient feed added on the platform schedule. Culture pH is controlled by base addition, and pCO2 is held at its level by the carbon dioxide fraction of the sparge gas. Temperature is controlled by the vessel jacket, and dissolved oxygen by sparging and agitation. Basal medium and feed are prepared under the media and feed procedure SOP-2004 from the platform formulations. Each factor of a design is held at the level that the design gives for the run (Appendices A and B), and every parameter that is not under study in a run is held at its set-point (Table 6.1). The culture is harvested at the culture duration of the run.

Inputs that are controlled to a platform specification are tested for identity and quality on receipt and are not characterized in this study. They include the medium and feed lots, the gases and the base. Their specifications were set on the platform and are unchanged for A-Mab, so the variation they contribute is bounded by the specification and by the release testing of each lot. Where a lot of medium or feed is used in more

than one design, its identity is recorded with the run, so that a lot effect can be examined if the data suggest one.

5.3 Analytical methods

Table 5.1 lists the 5 validated methods that will generate the study data. The released N-glycan map by 2-AB HILIC-UPLC measures afucosylation, galactosylation and high mannose from a single preparation, so the three glycan responses of any one run are not independent measurements. Size exclusion chromatography (SEC) measures aggregate as high molecular weight species (HMW). Imaged capillary isoelectric focusing (icIEF) measures acidic variants. The host cell protein ELISA and the residual DNA qPCR measure the two impurity attributes on the clarified harvest.

Table 5.1: Validated analytical methods and the attributes they measure.

Method	Procedure	Validation	Attributes	Measured on
N-Glycan Map (2-AB HILIC-UPLC)	SOP-3010	AMV-3010	Afucosylation, galactosylation and high mannose	every run
Charge Variants (icIEF)	SOP-3013	AMV-3013	Acidic variants	every run
Size-Variants (SEC-HPLC)	SOP-3011	AMV-3011	Aggregate	every run
Host-Cell Protein ELISA	SOP-3012	AMV-3012	Host cell protein	clarified harvest of every run
Residual DNA (qPCR)	SOP-3014	AMV-3014	Residual DNA	clarified harvest of every run

Table 5.2 gives the method for each size variant in the two designs.

Table 5.2: Methods used for size variants in the designed studies, production bioreactor.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	
Fragments, LMW	not in scope[a)]	

[a)] Fragments were not an attribute of this study.

Each method is used within its validated range and under its current validation report, and no method is modified for this study. Method precision enters the study twice. It contributes to the variation between the centre point replicates, and so to the pure

error used to test lack of fit. It also sets a floor under the smallest change in a response that the study can resolve, so an effect of the same size as the method precision is reported as detected but not distinguishable from analytical variation.

5.4 Sampling plan

Each run is sampled at harvest, and the harvest sample is assayed for the 5 modelled responses. The clarified harvest of each run is assayed for host cell protein and residual DNA. Samples are taken by the procedure in SOP-2003 and are handled and stored in the same way across runs, because a difference in hold time before assay would appear in the data as a run effect.

The culture is monitored through the run for viable cell density, viability and the principal metabolites, under SOP-2003. These measurements describe how the culture progressed and support the investigation of any run that fails an operating criterion. They are not responses of the designs, and no model is fitted to them.

The screening design includes 3 centre point runs and the response-surface design 4, all at the set-points in Table 6.1. The centre points are spread through the execution order, so that any drift in the medium and feed lots or in the analytical methods over the campaign enters the pure error.

5.5 Statistical methods

The designs are generated, randomized and analysed under the design of experiments procedure SOP-1002, with the methods of the master plan PCMP-001 (PCMP-001 §6) (Montgomery 2017). Each factor is coded so that the set-point is 0 and the ends of the range studied are -1 and $+1$. A coefficient then measures the change in the response from the set-point to the end of the range, and coefficients for different factors can be compared directly (CMC Biotech Working Group 2009). An effect is reported as twice the coded coefficient, which is the change in the response across the full range studied of that factor. A model term is significant when its p-value is below $\alpha = 0.05$.

The screening data are fitted by least squares to a model of the 5 main effects and the 10 two-factor interactions. The model has 16 terms and the design has 19 runs, which leaves 1 degree of freedom for lack of fit against 2 for pure error. With so few residual degrees of freedom, R^2 describes the fit to these runs but overstates how well the model would predict new runs. The screening analysis is therefore read as identifying which factors and which interactions are active. It is not used to predict a response, to define an operating region or to set a range (PCMP-001 §6.2).

The response-surface data are fitted to the full quadratic model in the 4 factors carried forward, which has 15 terms in 28 runs. The lack of fit test uses the 4 centre points and has 10 and 3 degrees of freedom. Each model is reported with R^2, adjusted R^2, predicted R^2 from the prediction error sum of squares (PRESS), the model F-test and the analysis of variance that splits the residual into lack of fit and pure error.

The residuals of every modelled response are examined against the predicted values and against run order, on a normal quantile plot, and as predicted against measured values. The response-surface model is the predictive model of this step, and every prediction, every PAR and the operating region come from it.

A factor with no term significant at $\alpha = 0.05$ for a response is reported as showing no detectable effect on that response over the range studied. That statement holds over the ranges studied and no further. It also holds only down to the smallest effect that the design and the analytical method can resolve, which the centre points measure directly. An effect that is significant but small is reported with the predicted change across the range beside it, so that statistical and practical significance are kept apart (CMC Biotech Working Group 2009).

A response-surface model predicts the mean response. At the edge of a region drawn on mean predictions, a single batch meets its criterion only about half the time if the variation is symmetrical (CMC Biotech Working Group 2009). For this reason each quality attribute is judged in the PAR analysis, in the scale-down qualification and in the operating region on the 95 % prediction interval of a single batch, not on the mean prediction (PCMP-001 §6.2).

Capability at commercial scale is estimated by simulating 2,000 batches (PCMP-001 §6.5). Each simulated batch draws the parameters of the response-surface models about their set-points, as in the PAR analysis (Section 8), and evaluates the fitted models together with the variation observed in the study. Cpk is computed against the drug substance criterion. The criteria of afucosylation, galactosylation, high mannose and acidic variants have two limits, and the Cpk of each is the lower of the two one-sided indices. The same simulation gives, for each attribute, the probability that the drug substance falls outside its criterion. The variation drawn on is that of the scale-down model, which does not carry the contribution of the commercial vessel to batch-to-batch variation, so the capability will be confirmed against the commercial batches of Stage 2.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists the 9 parameters of the step with the set-point, the range studied, the NOR and the study type of each. The set-point is the target at which the commercial process will be operated. The NOR is the band within which routine operation is expected to hold the parameter. The range studied is the knowledge space of this characterization. This plan assigns no parameter class. Classification is a result of the study, and the bioreactor report PCR-003 states the class of each parameter under the parameter classification procedure SOP-4001.

Table 6.1: Process parameters of the production bioreactor, with set-point, range studied, NOR and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
CO2 partial pressure (pCO2)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm/kg	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% air saturation	50	30–70	40–60	univariate
Initial viable cell conc.	10^6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	× standard	1.2	0.8–1.6	1–1.4	univariate
Nutrient feed-1 volume	% of working volume	12	9–15	10–14	univariate

In the designs each factor carries a code letter and is run at coded levels. Table 6.2 gives the natural value of each coded level and the designs in which each factor is varied. The codes are assigned in screening order and kept for the response-surface terms, so that a factor carries the same letter throughout the study.

Table 6.2: Factors of the designs, with their code letters and the natural value of each coded level.

Code	Factor	Unit	−1	0	+1	Designs
A	Culture pH	pH	6.6	6.85	7.1	screening and response surface
B	Culture temperature	°C	34	35	36	screening and response surface
C	CO2 partial pressure (pCO2)	mmHg	40	100	160	screening and response surface
D	Osmolality	mOsm/kg	360	400	440	screening only
E	Culture duration	days	15	17	19	screening and response surface

6.2 Screening design

The screening study S-BRX-SCR varies the 5 multivariate parameters in a two-level fractional factorial design. The design is a resolution V half fraction of the full factorial, with 16 factorial runs and 3 centre points, 19 runs in all. At resolution V all 5 main effects and all 10 two-factor interactions are estimable clear of one another, and each is aliased only with interactions of three factors or more. Appendix A gives the planned design. The runs are made in randomized order under SOP-1002, which protects the estimates against any drift in the model, the feed lots or the analytical methods over the execution period.

The design carries no quadratic term, so it cannot describe curvature. A difference between the mean of the centre points and the mean of the factorial runs is read as evidence of curvature in that response, and the response-surface design resolves it. A three-factor interaction would mean that the amount by which one parameter changes the effect of another itself changes with a third. No such behaviour has been seen on the platform, and the parameters carried into the response-surface design are re-estimated there in a design without such aliases.

pCO2 and osmolality are separate factors of the screening design. The design therefore estimates the main effect of each and their two-factor interaction. In routine manufacture a higher pCO2 brings a higher osmolality with it (Section 4.1), so the combined effect of the two at manufacturing scale includes the interaction that the screening estimates. This plan does not state how osmolality is brought to its design level in a run in which pCO2 also varies, and the report states how it was done.

6.3 Response-surface design

The response-surface study S-BRX-RSM varies 4 of the 5 multivariate parameters in a face-centred central composite design. The design has 16 factorial runs, 8 axial

runs at the centres of the faces of the design cube, and 4 centre points, 28 runs in all. The axial runs sit at the ends of the ranges studied. A rotatable design would place them beyond those ends, at a culture pH and a temperature that the vessel control holds only with difficulty and at which the culture no longer represents commercial operation. Each factor is therefore run at 3 levels. Appendix B gives the planned design, and the design supports the full quadratic model of Section 5.5.

The 4 parameters planned for this design are culture pH, culture temperature, culture duration and CO2 partial pressure (pCO2). The parameter not carried forward is osmolality, whose range studied is the narrowest of the 5 relative to its NOR, at 1.6 times the width of the NOR. Its main effect and its interactions with the other 4 are estimated in the screening design, and it is held at its set-point in every run of the response-surface design.

The subset above is the planned subset. The parameters actually carried into the response-surface design will be those that the screening analysis identifies as active on the attributes of Table 4.1. If screening identifies a different subset, the design will be amended before execution, the amendment will be approved by the functions listed in the approval record, and it will be reported in PCR-003 with the screening result that prompted it.

6.4 Univariate assessment

The univariate study S-BRX-UNI assesses dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume one factor at a time. Each parameter is run at the low end and the high end of its range studied, with every other parameter of Table 6.1 at its set-point, and the harvest is assayed for the 5 responses and for the two impurity attributes.

This design cannot determine whether the effect of one of these parameters depends on the level of another parameter (International Council for Harmonisation 2009). The range it supports for each parameter is therefore conditioned on the other parameters being at their set-points, and the report will state that condition wherever it states the range. The condition is accepted for these 4 parameters for the reason given in Section 4.3. If a univariate assessment shows an effect on a quality attribute that this mechanism does not account for, the parameter will be carried into a multivariate study and the plan amended.

7 Acceptance and decision criteria

The criteria below apply the acceptance criteria framework of the master plan PCMP-001 (PCMP-001 §7) to the production bioreactor. Any change to them after execution begins is recorded as an amendment to this plan (Section 13).

1. The scale-down model is qualified for an output when the conclusion for that output, whether it meets its criterion at all 81 combinations of NOR low, set-point and NOR high of the response-surface factors, is the same with the offset at either end of its 90 % confidence interval (Section 5.1). A model whose conclusion for an output changes between the two ends of the offset interval is not used for that output (PCMP-001 §7.1).
2. A run is accepted into the analysis when it was executed within the tolerances of SOP-2003 for the parameters not being varied, and when its harvest samples met the acceptance criteria of the analytical methods. A run that fails either condition is recorded as a deviation, investigated, and either repeated or dispositioned with a documented impact assessment. A study whose design is invalidated by a deviation is re-executed in full on material that meets the input specifications of the step, and the invalidated data set is retained and referenced in PCR-003 without being analysed.
3. A response-surface model is adequate for prediction when all of the following hold (PCMP-001 §7.2).

   a. The model F-test is significant at $\alpha = 0.05$.
   b. Lack of fit against pure error is not significant at $\alpha = 0.05$.
   c. Predicted R^2 falls short of R^2 by no more than 0.20.
   d. The residuals show no pattern against predicted values or run order, and no single run dominates the fit. A regression of the absolute residuals on the predicted values, and a regression of the residuals on run order, each show no slope significant at $\alpha = 0.05$. A run with a Cook’s distance above $4/n$ is examined by deleting it and repeating the judgement. The model fails this test only where a run’s Cook’s distance exceeds the median of the F distribution with p and $n - p$ degrees of freedom, where p is the number of model parameters and n the number of runs. The flag $4/n$ is a common screening cut-off for influential runs that scales with the number of runs, and it is 0.14 for the 28 runs of the response-surface design.
   e. The normal quantile plot of the residuals shows no departure from normality.

   A model that fails any of these is not used to define an operating region or to set a range. A model that fails the lack-of-fit test or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range (PCMP-001 §7.2). Where no adequate model results, the report states the response descriptively, against its criterion, over the runs of the designs.

4. An effect is significant when its term is significant at $\alpha = 0.05$. Significance alone does not make an effect important. An effect is practically important when the predicted change in the response across the range studied is large relative to the width of the criterion of that attribute. An effect that is significant but not practically important is reported as such, with both quantities given. Neither PCMP-001 nor SOP-4001 sets a numerical threshold for "large", so the report states the size of each effect against the width of the criterion and argues each judgement.
5. Each step output is judged against its intermediate acceptance criterion in Table 7.1, at manufacturing scale, by the analyses of Section 8. A quality attribute is judged on the 95 % prediction interval of a single batch. Afucosylation, galactosylation and high mannose keep their drug substance criteria, because no downstream step changes them. The criteria of afucosylation, galactosylation, high mannose and acidic variants are judged at both limits, and the criterion for aggregate at its upper limit.
6. An operating region is declared over the response-surface parameters where every modelled attribute meets its criterion at both ends of the 95 % prediction interval of a single run from the response-surface models (PCMP-001 §6.2). The region is declared only over the parameters varied in the response-surface design and only within the ranges studied. The NORs must lie inside the declared region. If they do not, the operating region is reported as it is found and the NOR is narrowed until it fits inside.
7. A proven acceptable range is established for each parameter against each attribute whose fitted model contains a significant term in that parameter, by the analyses of Section 8. A parameter has a PAR for each such attribute, and the range carried into the control strategy is their intersection (PCMP-001 §7.6). Where the analysis with the other factors varied across their NORs does not meet the criterion even with the parameter at its set-point, the report states that no PAR exists for the attribute and states the consequence for the NOR. The plan is met for a parameter when such a range is established and the NOR lies inside it for every one of those attributes.
8. The range of a univariate parameter is supported by its own data, with the other parameters at their set-points (PCMP-001 §4). A univariate result is judged by the rule of Section 6.4. An effect on a quality attribute that the mechanism does not account for carries the parameter into a multivariate study.
9. Each parameter will be classified under SOP-4001 from the effects demonstrated in these studies and from the risk that routine operation moves it outside the region in which those effects are acceptable (CMC Biotech Working Group 2009). A parameter whose variability affects a critical quality attribute is a critical process parameter or a well controlled critical process parameter, and the distinction between the two rests on the risk of falling outside the design space rather than on the size of the effect. A parameter that affects process performance without affecting a quality attribute is a key process parameter. A parameter with no demonstrated effect on either is a general process parameter. PCMP-001 Version 2.0 adds two rules to this one. An attribute is affected by a parameter when a term that contains the parameter is significant at $\alpha = 0.05$ (PCMP-001 §6.2). A significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical (PCMP-001 §7.7). PCMP-001

and SOP-4001 set no numerical threshold for a high risk, a practically relevant effect or a wide PAR. The report therefore states the size of each effect across the PAR beside its significance and the margin between each PAR and its NOR, and it argues each class on them.

10. Capability is judged in the master report PCMR-001 against two values that PCMP-001 §7.9 sets. A Cpk of at least 1.33 is expected of a critical quality attribute in routine manufacture. A Cpk below 1 requires an action in the control strategy. A Cpk between the two is acceptable, and the master report then states the side of the criterion that limits it and the parameters that drive it. The report of this step states the Cpk of each attribute with its margin to the limit and names the attribute whose capability is tightest.
11. The study is complete when the qualification of the scale-down model is decided for every output with a criterion, when an adequate model exists for every modelled response or the reason for its absence is reported, when an operating region containing the NORs is defined, when a PAR is established for every parameter and every attribute that its model shows it changes, and when every parameter of Table 6.1 is classified.

Table 7.1 gives the intermediate acceptance criteria of the step and their derivation. Each drug substance criterion is carried back through the worst case of every downstream step, with no margin (PCMP-001 §7.3). The worst case is the platform value from X-Mab, Y-Mab and Z-Mab, unless the verification described below replaced it with an A-Mab value. The aggregate criterion is carried back through the aggregate clearance factor of cation exchange and the aggregate increase across low-pH viral inactivation. It is the drug substance limit multiplied by the clearance factor, less the increase. From the unrounded values in force this gives $5.00 \times 1.4038 - 0.5586 = 6.46$ % HMW, against $5.00 \times 2.00 - 0.70 = 9.30$ % from the platform values alone. The upper limit for acidic variants is carried back through the platform worst case of the increase across low-pH viral inactivation, $40.0 - 1.00 = 39.0$ %. The lower limit is carried back unchanged at 18.0 %, which assumes that no downstream step lowers acidic variants (Section 12). Each increase is in percentage points of the attribute.

Table 7.1: Intermediate acceptance criteria for the production bioreactor output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Aggregates, HMW (%)	≤ 5.00	1.40	0.56	≤ 6.46	Carried back, tightened[a)] and widened[b)]
Acidic variants (%)	18.0–40.0	1.00	1.00	18.0–39.0	Carried back
Afucosylation (%)	2.00–13.0	1.00	0.00	2.00–13.0	DS criterion[c)]
Galactosylation (%)	10.0–40.0	1.00	0.00	10.0–40.0	DS criterion[c)]
High mannose (%)	3.00–10.0	1.00	0.00	3.00–10.0	DS criterion[c)]

[a)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[b)] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[c)] No downstream step changes the attribute, so the drug-substance criterion applies.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

Each criterion in force assumes that every downstream step performs at least as well as the worst case from which the criterion was derived. The report of each downstream step checks this from its own response-surface model, and the verification rule of PCMP-001 §7.5 lets the check change a criterion in either direction. Where the A-Mab worst case predicted within the NOR of a downstream step is worse than the platform worst case, the one-sided 95 % prediction bound of a single batch there replaces the platform value, and every criterion upstream becomes stricter. Where that bound is better than the platform value, and the response-surface model behind it meets the first three adequacy criteria of PCMP-001 §7.2 (a significant model, no significant lack of fit and the limit on overfitting) and is not an exact fit, the bound replaces the platform value, and every criterion upstream becomes wider. The fourth criterion, on the residuals and on influential runs, does not decide the change. The report that gives the bound examines it and states whether the bound depends on a single run.

The verification rule changed the criterion of this step for aggregate. The cation exchange model predicted a lower aggregate clearance within its NOR than the platform worst case of 2.00-fold. The one-sided 95 % prediction bound of a single batch for that clearance, 1.40-fold, tightened the criterion (cation exchange report PCR-007). The bound for the aggregate increase across low-pH viral inactivation, 0.56 percentage points against 0.70 on the platform, widened it (low-pH viral inactivation report PCR-006). The aggregate criterion in force, ≤ 6.46 % HMW, is stricter than

the a priori criterion of ≤ 9.30 %. The bioreactor report PCR-003 argues the change by its risk, as PCMP-001 §7.5 requires.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a parameter over which the attribute in question is shown by supportive data to remain acceptable (International Council for Harmonisation 2009). A PAR is computed for each of the 5 modelled attributes and each of the 4 response-surface factors, 20 combinations in all (PCMP-001 §6.4). Each PAR is computed from the response-surface model of the attribute, with the offset of the attribute subtracted. The criterion is the intermediate acceptance criterion in Table 7.1. Two analyses are run for each combination.

The first holds the other factors at the set-points in Table 6.1 and scans the parameter of interest across its full range studied, evaluating the fitted response-surface model at 201 grid points. The PAR at the set-points of the other factors is the contiguous range around the set-point over which the mean prediction meets the criterion. This analysis shows what the parameter does on its own.

The second analysis varies the other factors across their NORs, and PCMP-001 calls its result the NOR-propagated PAR. At each of 81 grid points across the range studied of the parameter of interest, 2,000 sets of settings are drawn for the other factors. Each factor is drawn from a normal distribution centred on its set-point, with a standard deviation equal to the width of its NOR divided by 6, so that 99.7 % of draws fall within the NOR. The draws are clipped to the range studied. Each simulated batch adds a residual drawn from a normal distribution with the residual standard deviation of the model, and it takes its own set of coefficients, drawn from the multivariate normal distribution given by the fitted coefficients and their covariance matrix. The 95 % prediction interval of a single batch runs from the 2.5th to the 97.5th percentile of the simulated batches. For an upper limit the 97.5th percentile must not exceed the limit, for a lower limit the 2.5th percentile must not fall below it, and for the two-sided criteria of afucosylation, galactosylation, high mannose and acidic variants both conditions apply. This analysis lets the other factors move as they move in routine operation, so the interaction terms of the fitted model contribute at every grid point.

In both analyses the PAR is the contiguous interval of the parameter, containing the set-point, over which the criterion is met. An interval that is met at the set-point but broken elsewhere in the range studied is reported as the contiguous interval, and the break is described where it occurs. Where the criterion is met across the whole range studied, the PAR is that range, and its edges are the edges of the study. The PAR with the other factors varied across their NORs is the one that supports a claim. It is in general narrower than the PAR at the set-points, for two reasons. It is judged on the interval of a single batch, which includes the residual and the uncertainty of

the coefficients, while the other is judged on the mean prediction. It must also hold while the rest of the operating point varies.

The PAR of pCO2 is a range at the set-point of osmolality, because the response-surface model holds osmolality there. The offset brings the predicted level to manufacturing scale, but it does not carry the rise in osmolality that comes with a rise in pCO2 at manufacturing scale (Section 6.2).

The report will give both PARs for every combination in one table, with the range studied, the criterion and its basis. For each attribute the report will also plot the PAR of the factor with the largest main effect on that attribute, with the acceptable region shaded green and the set-point and the NOR marked. A PAR is univariate in the parameter it names. The operating region of Section 7 is the multivariate region over which every modelled attribute is acceptable at once, and it is not the product of the individual PARs. Where the two differ, the difference is the effect of the interactions, and the report will say so.

9 Data management and integrity

All data generated under this plan are recorded to the ALCOA+ principles. Run records are captured in the validated electronic batch record system under SOP-2003, and analytical results are reported from the validated methods of Table 5.1 with their audit trail intact. Raw instrument data are retained in the form in which the instrument produced them.

The designs are generated, randomized and analysed under SOP-1002. The design file, the raw response data and the analysis script for each design are retained together, so that every table and figure in PCR-003 can be traced back to the runs that produced it. Any exclusion of a run from an analysis is documented with its justification before the analysis is run again.

Instruments used to set or measure a studied parameter have their calibration verified before execution and confirmed after it. A calibration failure found after execution is treated as a deviation, and the impact on the runs performed since the last successful verification is assessed and reported. Each study record states its execution window and the calibration status of the pCO2 probe that it used, which the preventive action PA3 of the QA investigation INV-001 requires of every study record.

All data and analyses are subject to a second-person review before they are used in PCR-003. The reviewer confirms that the recorded settings match the design, that the reported results match the raw data, and that the analysis follows Section 5.5.

10 Roles and responsibilities

Table 10.1 gives the responsibility of each function for this study. Process Development / MSAT writes the plan, Analytical Development, Manufacturing Science and Statistics review it, and Quality Assurance approves it (Section 13).

Table 10.1: Responsibilities of the functions that author, review and approve this plan.

Function	Responsibility
Process Development / MSAT	Author. Owns this plan, runs the studies on the scale-down model, analyses the data and writes PCR-003.
Analytical Development	Reviewer. Provides the validated methods, releases the analytical data, and assesses method performance where a result falls near a criterion.
Manufacturing Science	Reviewer. Confirms that the scale-down conditions and the set-points represent commercial operation, and reviews the operating region against the control capability of the commercial equipment.
Statistics	Reviewer. Approves the designs before execution, performs or reviews the model fitting and the capability simulation, and confirms model adequacy against Section 7.
Quality Assurance	Approver. Approves this plan and PCR-003, and dispositions any deviation raised during execution.

11 Deliverables and schedule

The deliverable of this plan is the bioreactor report PCR-003. It will report the executed designs, the fitted models and their adequacy, the operating region, the PARs, the capability estimate, the class of every parameter in Table 6.1, and any deviation or discrepancy from the plan. PCR-003 rolls up into the master report PCMR-001.

The activities run in the following phases. The parameters of the response-surface design are chosen from the screening result, so the phases run in this order.

1. Qualification runs of the scale-down model against the manufacturing-scale batches, and the estimate of the offset of each output. The output is the qualification record, referenced in PCR-003.
2. Execution of the screening design. The output is the screening data set and the effect estimates.
3. Execution of the response-surface design. The output is the response-surface data set and the fitted models.
4. Execution of the univariate assessments. The output is the univariate data set.
5. Analysis, decision on the qualification of the scale-down model for each output, capability simulation and parameter classification. The outputs are the operating region, the PARs and the parameter classes.
6. Reporting. The output is PCR-003.

Table 11.1 records the execution window of each study, from the start of its first run to the end of its last run. Every study of this plan ran in the scale-down bioreactor system whose pCO2 probe is EQ-BRX-205, and every study window closed before the next calibration of that probe was due on 2026-06-14. Confirmation of the ranges at commercial scale is not a deliverable of this plan. It will be made during the PPQ campaign at the receiving site under PTP-001 (U.S. Food and Drug Administration 2011).

Table 11.1: Execution window of each study of this plan, with the pCO2 probe it used and the next calibration due date of that probe.

Study	Study type	Start	End	Equipment	Next calibration due
S-BRX-SDM	Scale-down model qualification	2025-08-11	2025-09-05	EQ-BRX-205	2026-06-14
S-BRX-SCR	Screening design	2025-09-08	2025-10-03		
S-BRX-UNI	Univariate assessment	2025-10-06	2025-10-31		
S-BRX-RSM	Response-surface design	2025-10-20	2025-12-12		

12 Risks and assumptions

The main risks to the study are the following.

1. The scale-down model may not represent the commercial vessel for an attribute it is used to claim. The qualification by offsets addresses that risk for the outputs it compares (Section 5.1). The risk that remains is that the model represents mean performance without the batch-to-batch variation of the commercial vessel, and it is managed by confirming the ranges at commercial scale in Stage 2.
2. pCO2 depends on scale and is measured by the pCO2 probe of the scale-down bioreactor system (EQ-BRX-205). A drift of the probe between calibrations would move the true pCO2 of a run away from its design level, while the controller holds the probe reading at that level. The verification of the probe calibration before and after execution, with the impact assessment that follows a failure, addresses this risk (Section 9).
3. Material supply. The studies need vials of the working cell bank and lots of basal medium and feed sufficient for both designs. A change of medium or feed lot within a design would add variation that the design does not model, so the lot is recorded with each run and lots are reserved for the campaign where the quantity allows.
4. Analytical precision. A response measured near the edge of its criterion carries the precision of its method into every conclusion drawn about it, and an effect smaller than that precision cannot be resolved by any number of runs at this design size. The report states the consequence wherever a range is set by a boundary that the method cannot resolve sharply.

The study rests on the following assumptions.

1. The platform experience of Section 4.1 transfers to A-Mab. This is supported by the shared cell line construction, medium, feed strategy and control scheme, and the qualification runs test it directly.
2. The parameter list in Table 6.1 is complete for this step, as the risk assessment RA-001 established. A univariate result that the mechanism does not account for would call it into question.
3. The attributes in Table 4.1 are the attributes that the culture conditions change, and the attributes that the purification train removes are characterized at the steps that remove them.
4. The offset between the scales, estimated at the set-points, holds across the range studied.
5. Interactions of three or more factors are negligible, as the screening design requires (Section 6.2).

6. The univariate parameters do not change the effect of the other parameters within their ranges, because the feed holds the nucleotide sugar precursors in excess (Section 4.3), and their effects do not reverse between the two ends of each range studied.
7. No downstream step lowers acidic variants, so the drug substance lower limit applies to the output of this step.
8. In routine operation each parameter varies about its set-point as a normal distribution whose standard deviation is the width of its NOR divided by 6 (Section 8).

13 Approvals

This plan is written, reviewed and approved by the functions listed in the approval record at the front of this document. Any change to the designs, the ranges, the analyses or the acceptance criteria after approval is documented as an amendment, approved by the same functions, and reported in PCR-003 with the reason for the change. Version 2.0 takes effect on 2026-09-30.

Appendix A. Planned screening design

Table A.1 gives the coded settings of the 19 planned screening runs, in the code letters of Table 6.2. The order shown is the design order, and execution follows the randomized order generated under SOP-1002.

Table A.1: Planned screening design of study S-BRX-SCR in coded units.

Run type	Run	A	B	C	D	E
Factorial	1	−1	−1	−1	−1	+1
	2	−1	−1	−1	+1	−1
	3	−1	−1	+1	−1	−1
	4	−1	−1	+1	+1	+1
	5	−1	+1	−1	−1	−1
	6	−1	+1	−1	+1	+1
	7	−1	+1	+1	−1	+1
	8	−1	+1	+1	+1	−1
	9	+1	−1	−1	−1	−1
	10	+1	−1	−1	+1	+1
	11	+1	−1	+1	−1	+1
	12	+1	−1	+1	+1	−1
	13	+1	+1	−1	−1	+1
	14	+1	+1	−1	+1	−1
	15	+1	+1	+1	−1	−1
	16	+1	+1	+1	+1	+1
Centre	17	0	0	0	0	0
	18	0	0	0	0	0
	19	0	0	0	0	0

Appendix B. Planned response-surface design

Table B.1 gives the coded settings of the 28 planned response-surface runs, in the code letters of Table 6.2. The columns follow the order in which the axial runs vary the factors. The axial runs are set at −1 and +1, which makes the design face-centred. Osmolality, code D, is not a factor of this design and is held at its set-point in every run. The order shown is the design order, and execution follows the randomized order generated under SOP-1002.

Table B.1: Planned response-surface design of study S-BRX-RSM in coded units.

Run type	Run	A	B	E	C
Factorial	1	−1	−1	−1	−1
	2	−1	−1	−1	+1
	3	−1	−1	+1	−1
	4	−1	−1	+1	+1
	5	−1	+1	−1	−1
	6	−1	+1	−1	+1
	7	−1	+1	+1	−1
	8	−1	+1	+1	+1
	9	+1	−1	−1	−1
	10	+1	−1	−1	+1
	11	+1	−1	+1	−1
	12	+1	−1	+1	+1
	13	+1	+1	−1	−1
	14	+1	+1	−1	+1
	15	+1	+1	+1	−1
	16	+1	+1	+1	+1
Axial	17	−1	0	0	0
	18	+1	0	0	0
	19	0	−1	0	0
	20	0	+1	0	0
	21	0	0	−1	0
	22	0	0	+1	0
	23	0	0	0	−1
	24	0	0	0	+1
Centre	25	0	0	0	0
	26	0	0	0	0
	27	0	0	0	0
	28	0	0	0	0

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.

International Council for Harmonisation. 2009. “ICH Q8(R2): Pharmaceutical Development.” ICH.

———. 2023. “ICH Q9(R1): Quality Risk Management.” ICH.

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. 9th ed. Hoboken, NJ: Wiley.

U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.